

ECON 381: Homework 2 Solutions

Fall 2026

Instructor answer key – 100 points total

Equivalent correct reasoning and algebra should receive credit.

Grading summary

Section	Points
I. Conceptual Questions	25
II. True or False	15
III. Analytical Questions	60
Total	100

I. Conceptual Questions (25 points)

1. Law of Large Numbers and Central Limit Theorem. [5 points]

(a) LLN. [2 points]

Suppose $Y_1, \dots, Y_n$ are independent and identically distributed with finite population mean $\mu_Y = E[Y_i]$. Then

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{p} \mu_Y.$$

Equivalently, for every $\varepsilon > 0$,

$$\Pr(|\bar{Y} - \mu_Y| > \varepsilon) \longrightarrow 0 \quad \text{as } n \longrightarrow \infty.$$

Thus, large discrepancies between the sample average and population mean become increasingly unlikely as the sample grows.

(b) CLT. [2 points]

Suppose in addition that Y_i has finite, positive variance. Then

$$\frac{\sqrt{n}(\bar{Y} - \mu_Y)}{\text{SD}(Y)} \xrightarrow{d} N(0, 1).$$

Therefore, for sufficiently large n ,

$$\bar{Y} \text{ is approximately } N\left(\mu_Y, \frac{\text{Var}(Y)}{n}\right).$$

(c) Different roles. [1 point]

The LLN establishes consistency: $\bar{Y}$ approaches μ_Y . The CLT describes the approximate shape and scale of the remaining sampling error, which makes standard errors, confidence intervals, and hypothesis tests possible.

2. **Why estimators are treated as random. [5 points]**

Solution. Before the sample is drawn, the observations that will enter the dataset are unknown. Because an estimator is a function of those random observations, a different random sample generally produces a different value of $\bar{Y}$ or $\hat{\beta}_1$. The fixed parameter does not change, but the estimator and its sampling error,

$$\hat{\theta} - \theta,$$

vary across repeated samples. The estimator's sampling distribution quantifies this variability and allows researchers to judge the precision and reliability of the one estimate they actually observe. After the data are collected, the reported estimate is a fixed number, but its unknown distance from the true parameter remains the central inferential concern.

3. **A 99% confidence interval and sample size. [5 points]**

- (a) **[2 points]** For a two-sided 99% interval, $\alpha = 0.01$, so each tail contains $\alpha/2 = 0.005$. The critical value is

$$\Phi^{-1}(0.995) = 2.5758 \approx 2.576 \quad (\text{or } 2.58).$$

Thus, the approximate interval is

$$\boxed{\bar{Y} \pm 2.58 \text{SE}(\bar{Y})}.$$

- (b) **[2 points]** For an i.i.d. sample average,

$$\text{SD}(\bar{Y}) = \frac{\text{SD}(Y)}{\sqrt{n}}, \quad \text{SE}(\bar{Y}) = \frac{s_Y}{\sqrt{n}}.$$

Holding dispersion approximately fixed, multiplying the sample size by k divides both the standard error and the confidence-interval width by $\sqrt{k}$. For example, quadrupling n approximately halves both quantities.

- (c) **[1 point]** No. A larger sample generally makes the interval narrower and the large-sample approximation more accurate, but it does not guarantee coverage in the realized sample. The fixed mean is either inside the realized interval or outside it. The confidence level describes the (long-run) success rate of the confidence-interval procedure: if we repeatedly drew samples and constructed an interval from each sample, the proportion of intervals containing the true parameter would approach the stated confidence level.

4. **Sample standard deviation versus standard error. [5 points]**

Solution. The sample standard deviation

$$s_Y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}$$

measures the dispersion of individual observations around the sample average and estimates the population standard deviation σ_Y . In contrast,

$$\text{SE}(\bar{Y}) = \frac{s_Y}{\sqrt{n}}$$

estimates the dispersion of sample averages across repeated samples of size n . Both have the same units as Y , but the standard error becomes smaller as n increases, whereas the population dispersion of individual outcomes need not shrink.

5. **Large covariance versus correlation. [5 points]**

Solution. No. Correlation standardizes covariance by both standard deviations:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$$

A large positive covariance may simply reflect large measurement units or large marginal dispersions. For example, replacing X by $X^* = 1,000,000X$ multiplies $\text{Cov}(X, Y)$ by 1,000,000, but it also multiplies $\text{SD}(X)$ by the same factor, leaving the correlation unchanged. Therefore, the magnitude of covariance alone does not show that correlation is close to one.

II. True or False (15 points)

Award one point for the correct label and up to two points for a correct explanation.

1. **False. [3 points]** The population parameter is fixed; it does not change across repeated samples. The interval is random before sampling. A 95% procedure means that approximately 95% of the intervals constructed from repeated samples contain the fixed parameter. Once an interval has been realized, it either contains the parameter or it does not.
2. **False. [3 points]** A standard error measures the sampling variability of an estimator, such as $\bar{Y}$, across repeated samples. The standard deviation of Y measures variability of the underlying variable in the population.
3. **True. [3 points]**

Assuming both variables have finite, positive standard deviations, $\text{Corr}(X, Y) = 1$ implies

$$Y = \mu_Y + \frac{\sigma_Y}{\sigma_X}(X - \mu_X) \quad \text{with probability one.}$$

The slope σ_Y/σ_X is positive, so realized sample observations lie on that positively sloped line, apart from probability-zero events or numerical rounding.

4. **False. [3 points]** The LLN is a probabilistic convergence result, not a monotonicity result. A particular larger sample can produce an average farther from the population mean even though large errors become less likely as n grows.
5. **False. [3 points]** For the same estimate and standard error, the 99% interval uses the larger critical value 2.58 rather than 1.96. It is therefore wider, because covering more of the standard-normal distribution requires extending farther into both tails.

III. Analytical Questions (60 points)

1. **A sample mean and its uncertainty. [30 points]**

The observations are 8, 10, 12, 14, 16, and $n = 5$.

(a) **[3 points]**

$$\bar{Y} = \frac{8 + 10 + 12 + 14 + 16}{5} = \frac{60}{5} = \boxed{12}.$$

(b) **[5 points]** The deviations and squared deviations are

Y_i	$Y_i - \bar{Y}$	$(Y_i - \bar{Y})^2$
8	-4	16
10	-2	4
12	0	0
14	2	4
16	4	16
		$\sum = 40$

Therefore,

$$s_Y^2 = \frac{40}{5 - 1} = \boxed{10}, \quad s_Y = \sqrt{10} = \boxed{3.1623 \text{ (approximately)}}.$$

(c) **[4 points]**

$$\text{SE}(\bar{Y}) = \frac{s_Y}{\sqrt{n}} = \frac{\sqrt{10}}{\sqrt{5}} = \sqrt{2} = \boxed{1.4142 \text{ (approximately)}}.$$

(d) **[5 points]** The margin of error is

$$1.96 \text{ SE}(\bar{Y}) = 1.96\sqrt{2} = 2.7719.$$

Hence,

$$\begin{aligned} \bar{Y} \pm 1.96 \text{ SE}(\bar{Y}) &= 12 \pm 2.7719 \\ &= [9.2281, 14.7719] \\ &\approx \boxed{[9.23, 14.77]}. \end{aligned}$$

(e) **[6 points]** We cannot determine whether this particular interval contains μ_Y , because the population mean is unknown. A 95% confidence level is a statement about the procedure: over repeated random samples, approximately 95% of intervals constructed by a valid 95% procedure contain the fixed population mean. It is not a claim that the fixed mean has a 95% probability of lying in this already realized interval.

With only five observations, the CLT need not provide a good normal approximation, especially if the population is non-normal (like skewed or heavy-tailed). Thus, the nominal 95% coverage of this normal-critical-value interval may be inaccurate.

(f) **[7 points]** If $n = 1000$ while $\bar{Y} = 12$ and $s_Y = \sqrt{10}$, then

$$\text{SE}(\bar{Y}) = \frac{\sqrt{10}}{\sqrt{1000}} = \sqrt{\frac{10}{1000}} = \boxed{0.10}.$$

The interval is

$$\begin{aligned} 12 \pm 1.96(0.10) &= 12 \pm 0.196 \\ &= [11.804, 12.196] \\ &\approx \boxed{[11.80, 12.20]}. \end{aligned}$$

Its width is 0.392, compared with approximately 5.5437 when $n = 5$. The width ratio is

$$\frac{0.392}{5.5437} = \sqrt{\frac{5}{1000}} \approx 0.0707.$$

Thus, the $n = 1000$ interval is only about 7.1% as wide (about 14.1 times narrower). This follows from the $1/\sqrt{n}$ relationship: because the sample size increases by a factor of 200, the width of the confidence interval shrinks by a factor of

$$\frac{1}{\sqrt{200}} \approx 0.0707 = 7.07\%.$$

The larger sample also makes the normal approximation more credible, but it still does not guarantee that the realized interval covers μ_Y .

2. Population and sample regression lines. **[30 points]**

The population model and fitted sample line are

$$Y_i = 2 + 1.5X_i + u_i, \quad \hat{Y}_i = 3 + X_i.$$

(a) **[6 points]** The population coefficients are

$$\boxed{\beta_0 = 2 \quad \text{and} \quad \beta_1 = 1.5}.$$

They are fixed features of the specified population relationship. The sample estimates are

$$\boxed{\hat{\beta}_0 = 3 \quad \text{and} \quad \hat{\beta}_1 = 1.0}.$$

Before sampling, these estimators are random and generally change when a different sample is drawn. After observing the current sample, their realized numerical values are fixed.

(b) **[12 points]** For each observation, compute

$$\beta_0 + \beta_1 X_i = 2 + 1.5X_i, \quad u_i = Y_i - (2 + 1.5X_i),$$

$$\hat{Y}_i = 3 + X_i, \quad \hat{u}_i = Y_i - (3 + X_i).$$

The completed table is

i	X_i	Y_i	$\beta_0 + \beta_1 X_i$	u_i	$\hat{Y}_i$	$\hat{u}_i$
1	1	5	3.5	1.5	4	1
2	2	4	5.0	-1.0	5	-1
3	3	5	6.5	-1.5	6	-1
4	4	8	8.0	0.0	7	1

(c) [6 points] The regression error

$$u_i = Y_i - (\beta_0 + \beta_1 X_i)$$

is the vertical distance from an observation to the *population* regression line. Researchers generally cannot compute it because the population coefficients are unknown. The residual

$$\hat{u}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$$

is the vertical distance from an observation to the *estimated sample* line. It is computable after OLS produces $\hat{\beta}_0$ and $\hat{\beta}_1$.

(d) [6 points] From the table, the residuals are 1, -1, -1, 1. Therefore,

$$\sum_{i=1}^4 \hat{u}_i = 1 - 1 - 1 + 1 = \boxed{0},$$

and

$$\begin{aligned} \sum_{i=1}^4 X_i \hat{u}_i &= 1(1) + 2(-1) + 3(-1) + 4(1) \\ &= 1 - 2 - 3 + 4 \\ &= \boxed{0}. \end{aligned}$$

As a cross-check, $\bar{X} = 2.5$ and $\bar{Y} = 5.5$, and

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{5}{5} = 1, \quad \hat{\beta}_0 = 5.5 - 1(2.5) = 3.$$

Thus, the stated fitted line is exactly the OLS line for these four observations.